

4.3.2.4. Operational Service Domain and Host Name

For link-local Multicast DNS the service domain SHALL be **local**. For Unicast DNS such as used on Thread the service domain SHALL be as configured automatically by the Thread Border Router.

For DNS-SD a target host name is required, in addition to the instance name. The target host name SHALL be constructed using one of the available link-layer addresses, such as a 48-bit device MAC address (for Ethernet and Wi-Fi) or a 64-bit MAC Extended Address (for Thread) expressed as a fixed-length twelve-character (or sixteen-character) hexadecimal string, encoded as ASCII (UTF-8) text using capital letters, e.g., **B75AFB458ECD.<domain>**. In the event that a device performs MAC address randomization for privacy, then the target host name SHALL use the privacy-preserving randomized version and the hostname SHALL be updated in the record every time the underlying link-layer address rotates. Note that it is legal to reuse the same hostname on more than one interface, even if the underlying link-layer address does not match the hostname on that interface, since the goal of using a link-layer address is to ensure local uniqueness of the generated hostname. If future link layers are supported by Matter that do not use 48-bit MAC addresses or 64-bit MAC Extended Address identifiers, then a similar rule will be defined for those technologies.

4.3.2.5. Operational Discovery Service Records

After discovery, IPv6 addresses are returned in the AAAA records and key/value pairs are returned in the DNS-SD TXT record. The TXT record MAY be omitted if no keys are defined.

Nodes SHALL publish AAAA records for all available IPv6 addresses upon which they are willing to accept operational messages.

Only the common TXT record key/value pairs defined in [Section 4.3.4, “Common TXT Key/Value Pairs”](#) are defined for use in Operational Discovery.

Nodes SHALL silently ignore TXT record keys that they do not recognize.

4.3.2.6. Performance Recommendations

To improve overall performance of operational discovery, especially in large installations, the following recommendations SHOULD be taken in account:

1. Nodes SHOULD cache the last-known IPv6 address and port for each peer Node with which they interact from their SRV record resolved using DNS-SD, to save the cost of a run-time network lookup operation when not needed. When the IPv6 address and port for a peer Node is not known, or an attempt to communicate with a peer Node at its last-known IPv6 address and port does not appear to be succeeding within the expected network round-trip time (i.e., the retransmission timeout value for the first message packet) a Node SHOULD then perform a run-time discovery in parallel, to determine whether the desired Node has acquired a new IPv6 address and/or port [[RFC 8305](#)].
2. Nodes SHOULD respond to nonspecific service enumeration queries for the generic Matter Operational Discovery service type (**_matter._tcp**), but these queries SHOULD NOT be used in routine operation, and instead it is RECOMMENDED that they only be used for diagnostics purposes or to determine new membership within a fabric. When used, it is RECOMMENDED that service enumeration employ the **_I<HHHH>** Fabric-specific subtype to only enumerate the desired